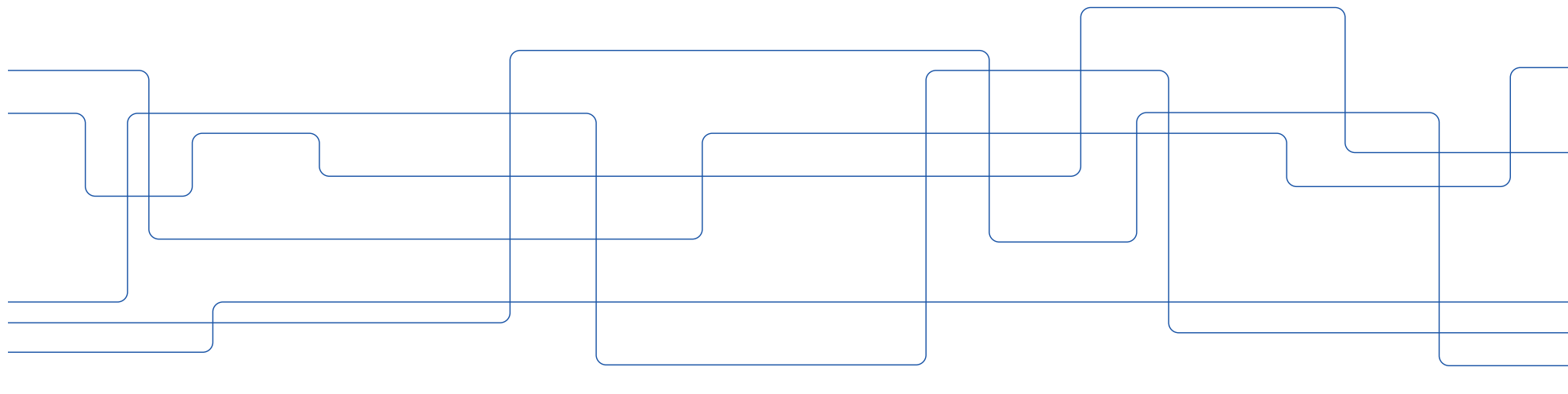




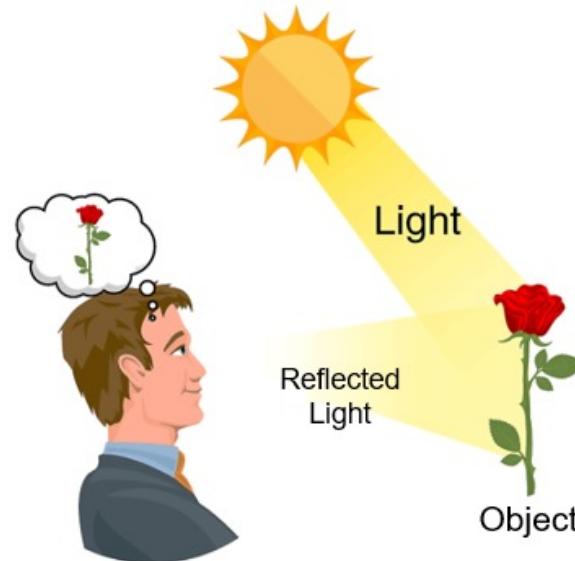
KTH ROYAL INSTITUTE
OF TECHNOLOGY

Introduction to Computer Vision

Mårten Björkman



What does it mean to see?

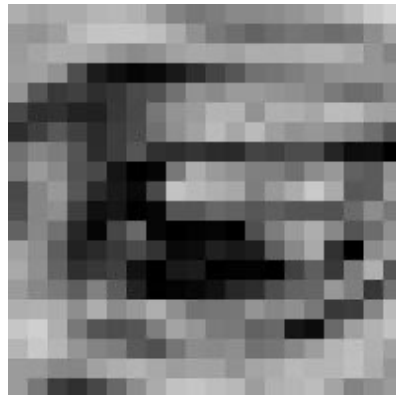


- We say that we 'see an object', but we really see light reflected on the object.

Reflected light = Incoming light + Object properties

- Vision is an 'active process' for deriving efficient symbolic representations of the world from the light reflected from it.

But what does a computer see?



174	164	172	171	181	183	177	157	131	119	125	137	137	141	147	157	169	184	200	211
151	147	159	183	196	197	197	190	171	144	121	110	99	109	117	114	109	117	142	164
155	165	174	172	164	148	138	145	161	168	166	161	173	172	173	179	180	170	149	132
158	161	146	78	28	15	6	10	25	46	69	86	113	117	126	136	145	149	149	147
147	86	57	64	46	54	74	82	113	146	138	154	154	148	144	137	122	108	112	134
72	46	49	35	33	56	73	119	148	156	180	179	146	176	174	166	187	186	163	159
46	63	88	64	43	61	75	112	144	162	187	167	186	158	146	151	145	132	112	83
112	134	124	84	44	61	85	48	45	53	44	42	68	64	76	76	37	13	15	0
122	154	138	82	72	21	0	5	120	173	154	144	128	127	112	140	152	97	91	145
73	134	153	87	41	25	10	74	194	184	175	168	126	153	172	201	171	87	91	156
78	134	143	63	16	22	1	6	85	85	103	97	87	94	84	84	82	106	141	120
98	135	114	39	5	70	105	6	0	3	5	0	79	125	154	173	136	84	108	144
160	142	94	33	42	57	74	51	9	12	32	24	29	95	137	169	49	0	140	158
169	144	100	47	55	64	18	0	25	28	0	3	4	0	65	95	56	124	176	84
154	136	122	94	129	92	34	1	4	8	17	39	45	78	99	102	152	126	55	113
174	177	171	142	173	139	153	174	143	115	118	119	148	134	143	120	56	60	134	185
198	207	175	120	84	77	85	126	163	150	123	121	98	105	21	12	106	159	178	193
180	200	181	149	132	125	90	80	117	140	137	142	113	118	135	157	180	190	176	161
155	154	136	134	161	174	180	168	157	162	161	145	165	171	185	180	181	191	173	149
155	119	61	48	71	104	141	171	193	197	189	184	157	177	181	188	164	131	143	155

- A computer just sees a bunch of numbers, organised in a two-dimensional array.
- For colour images you have, you have three values per pixel (red, green, blue).
- For video sequences you have an additional dimension corresponding to time.

Can computers match human vision?



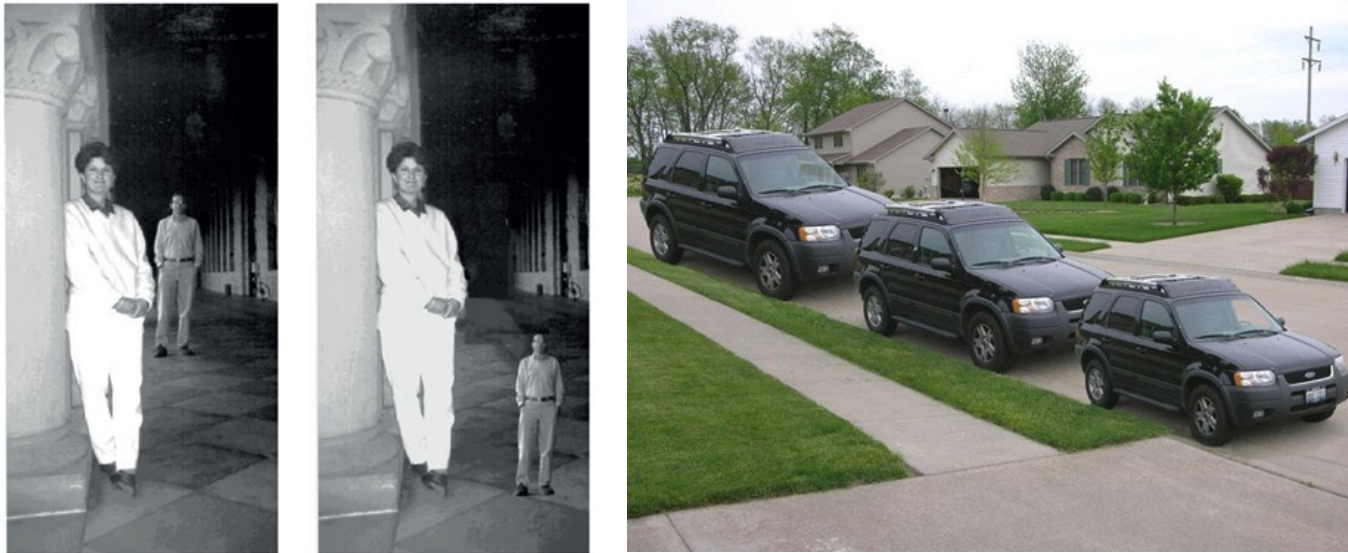
- Computers can be better at “easy” things (when things look the way they usually do)
- Humans are much better at “hard” things (when things are highly unusual)

But human vision has limitations - illusions



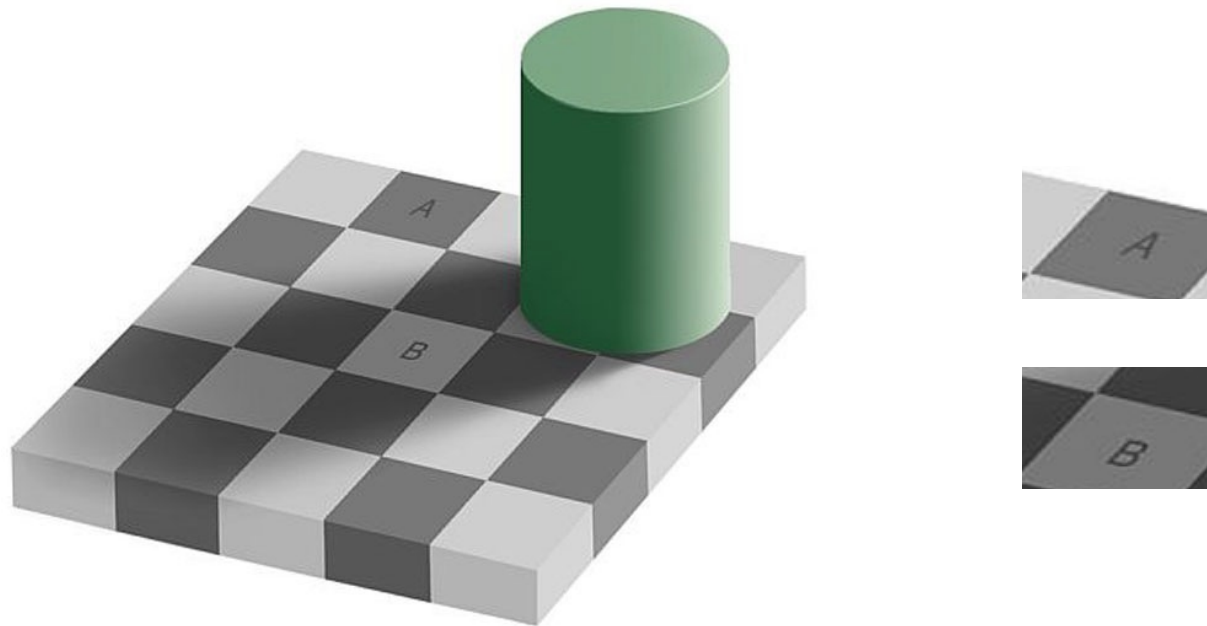
- If we look locally at an impossible object, it makes perfect sense, but globally it does not.
- We take multiple snapshots and integrate to a whole, i.e. vision is an *active process*.

Depth illusion – size constancy



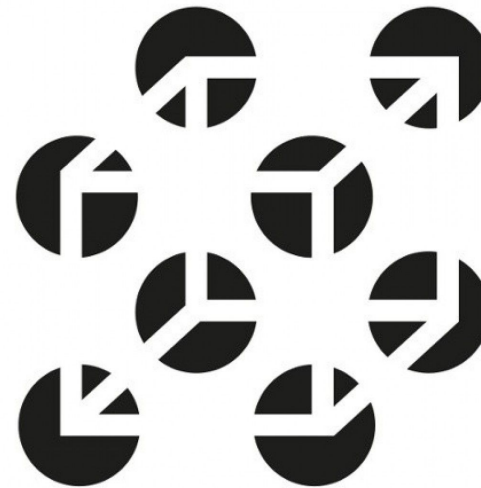
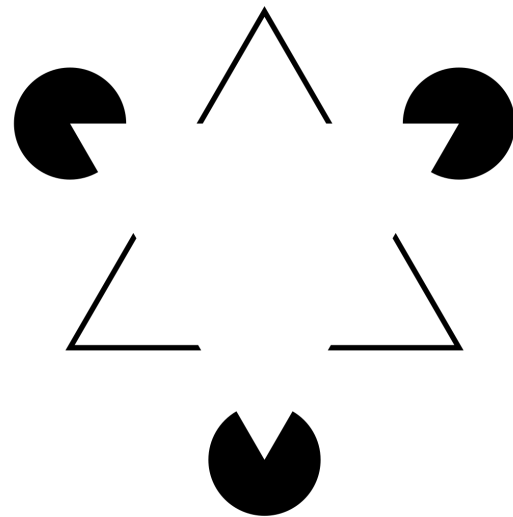
- Projective size = Object size + Viewing distance, but we tend to invert the problem and directly see the Object size.
- Context provides information on Viewing distance, which allows us to be fooled.

Color illusion – color constancy



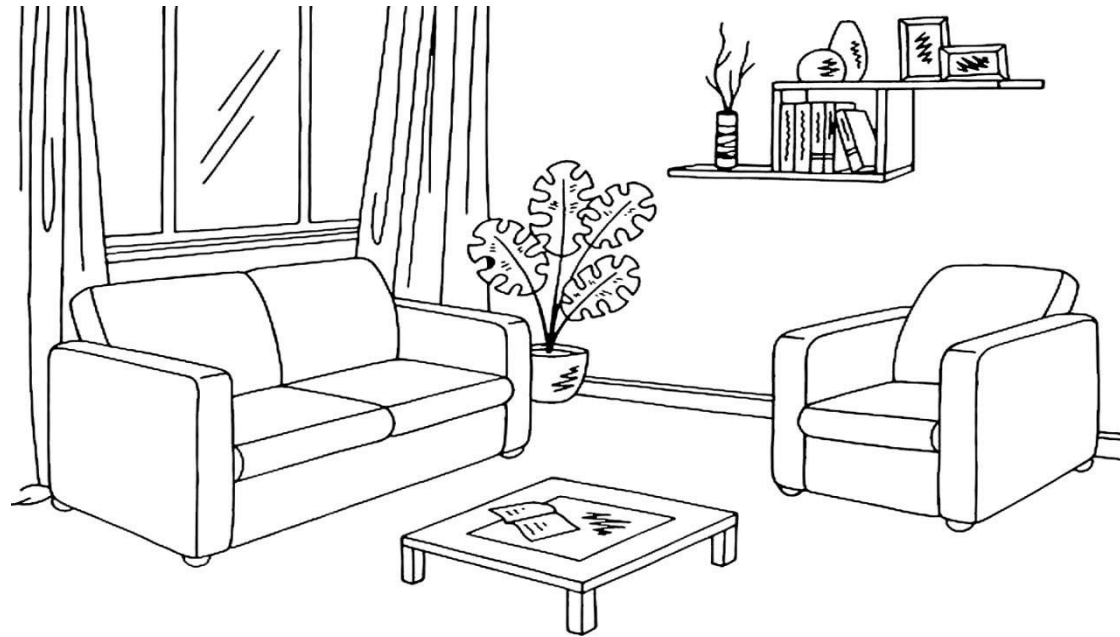
- Image color = Object color + Illumination, but (again) we tend to invert the problem and directly see the Object color.
- Context provides information on Illumination, which (again) allows us to be fooled.

Subjective contours



- Using prior experience, we tend to 'fill in the blanks' with information that is not there.
- In many cases our interpretation is correct, but in other cases it is not.

The importance of discontinuities



- We see discontinuities (edges) more than what is between the discontinuities.
- It is logical, since the colours are usually much less important for interpretation.
- Again, we fill in the blanks, without even knowing it.

Lessons learned from human vision



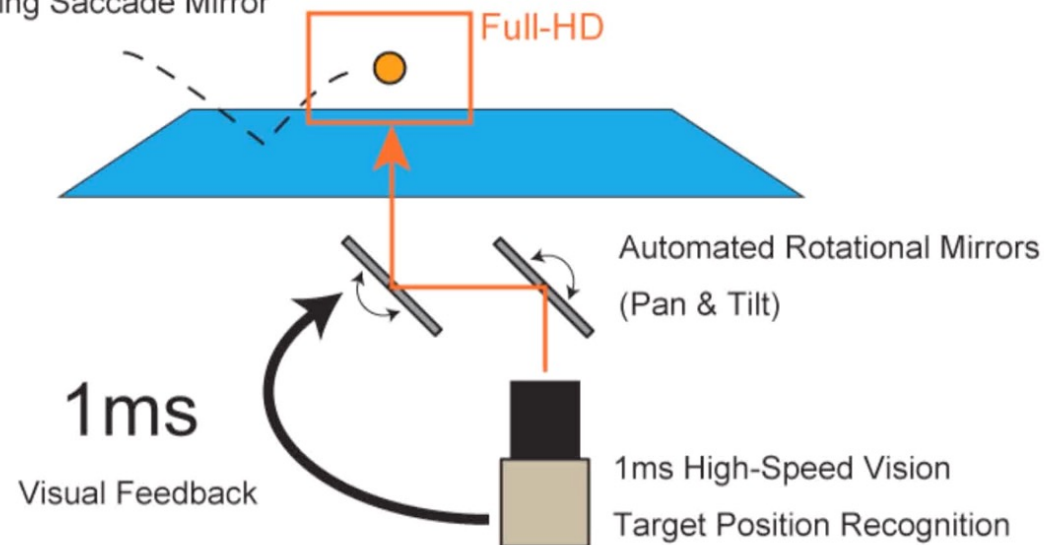
- Prior experience and context is very important for interpretation.
- Keep all possible interpretation in mind and update over time.
- Finding and interpreting edges should in many cases be enough.
- Our own vision system leads us to believe that it is easier than it really is.

A 1000 Hz video camera

1ms Auto Pan-Tilt

using Saccade Mirror

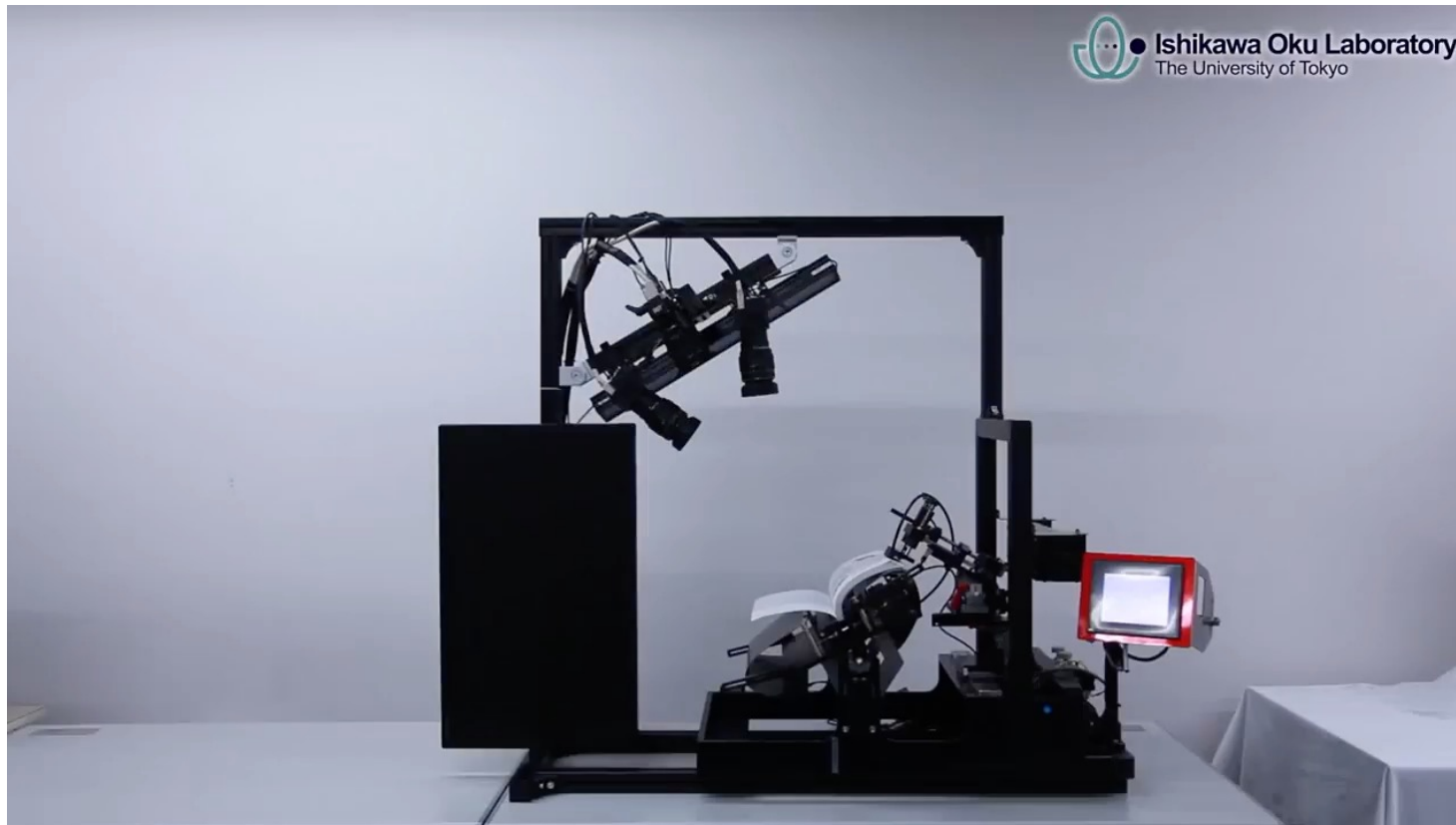
 **Ishikawa Oku Laboratory**
The University of Tokyo



- Most video cameras run in 25-60 Hz, since we cannot observe faster changes anyway.
- But for machine vision, we do not need such a limitation. Why not 1000 Hz?



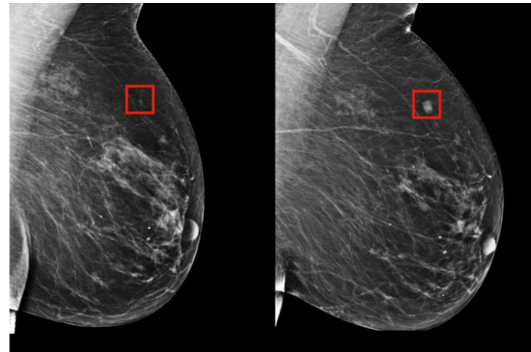
Automatic book scanning



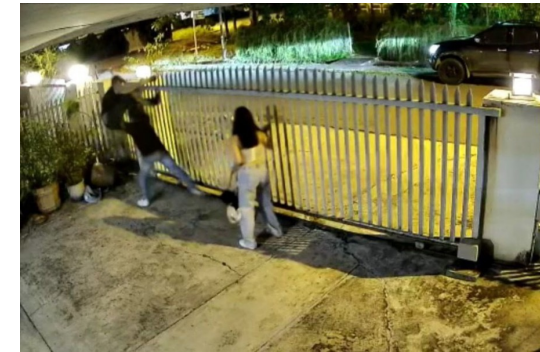
Why is computer vision relevant?



Safety



Health



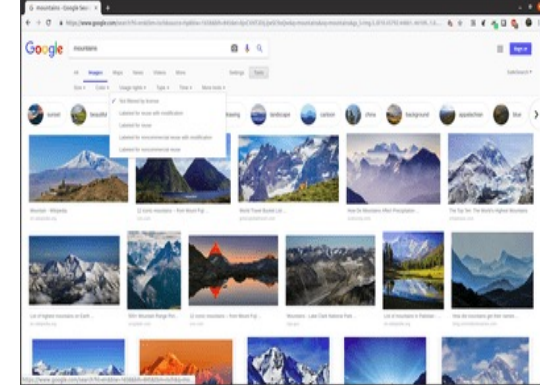
Surveillance



Comfort



Entertainment



Access

Some companies from people at RPL



13th Lab (Facebook)



Tracab (Chyron Hego)



Volumental

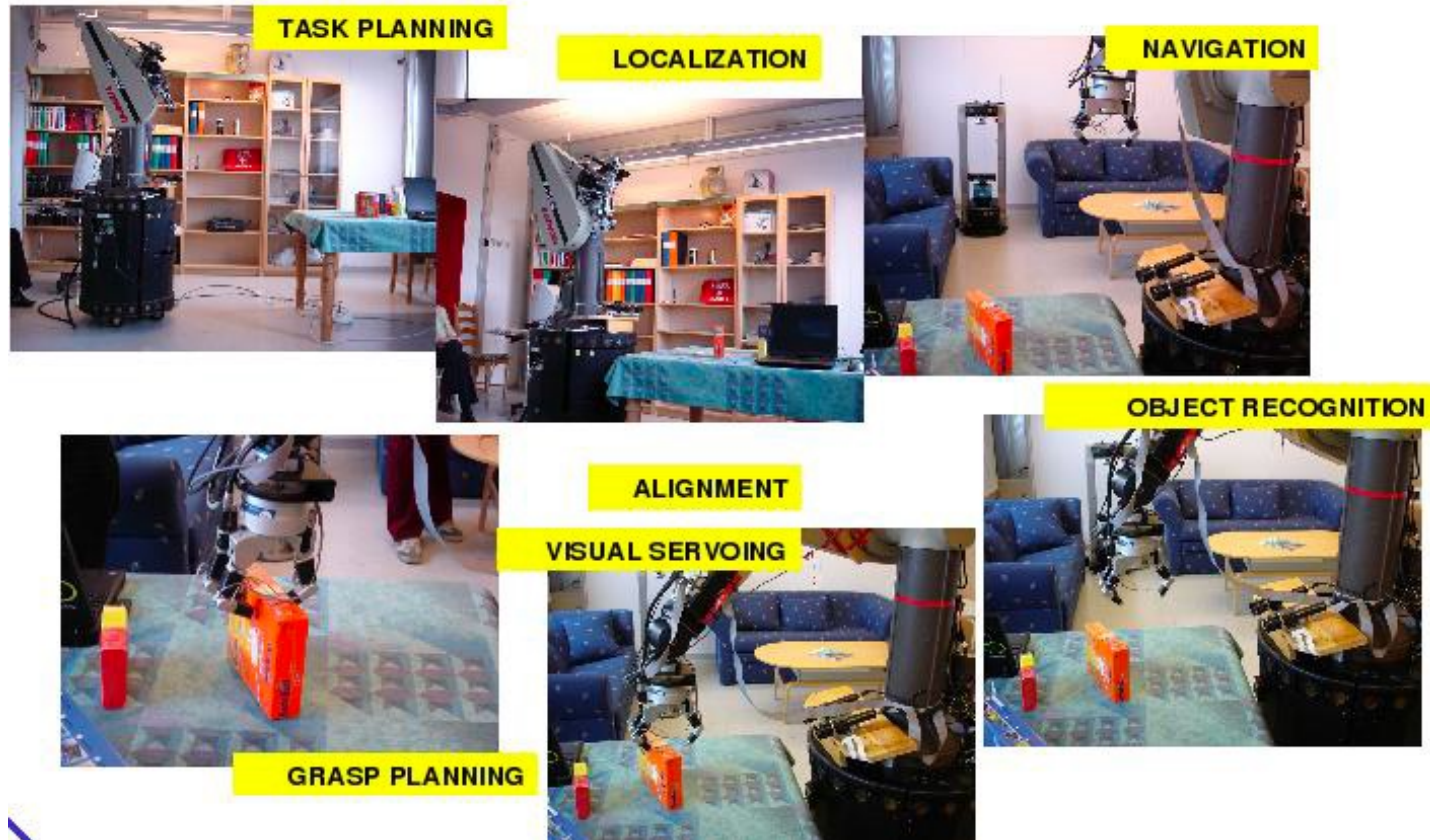


Univrse



Gleechi

Example: Robotic tasks using computer vision



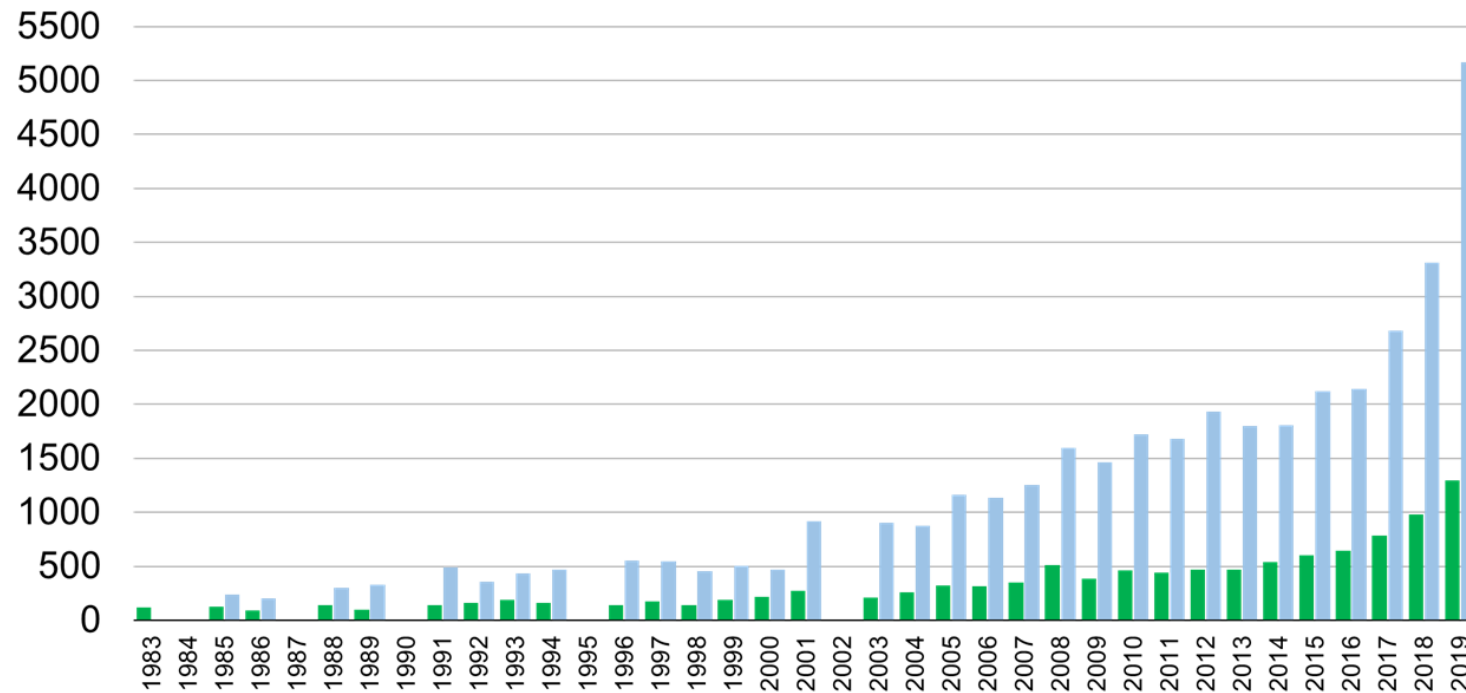
From experiments done at RPL in 2004, equally relevant today.



Why is computer vision interesting?

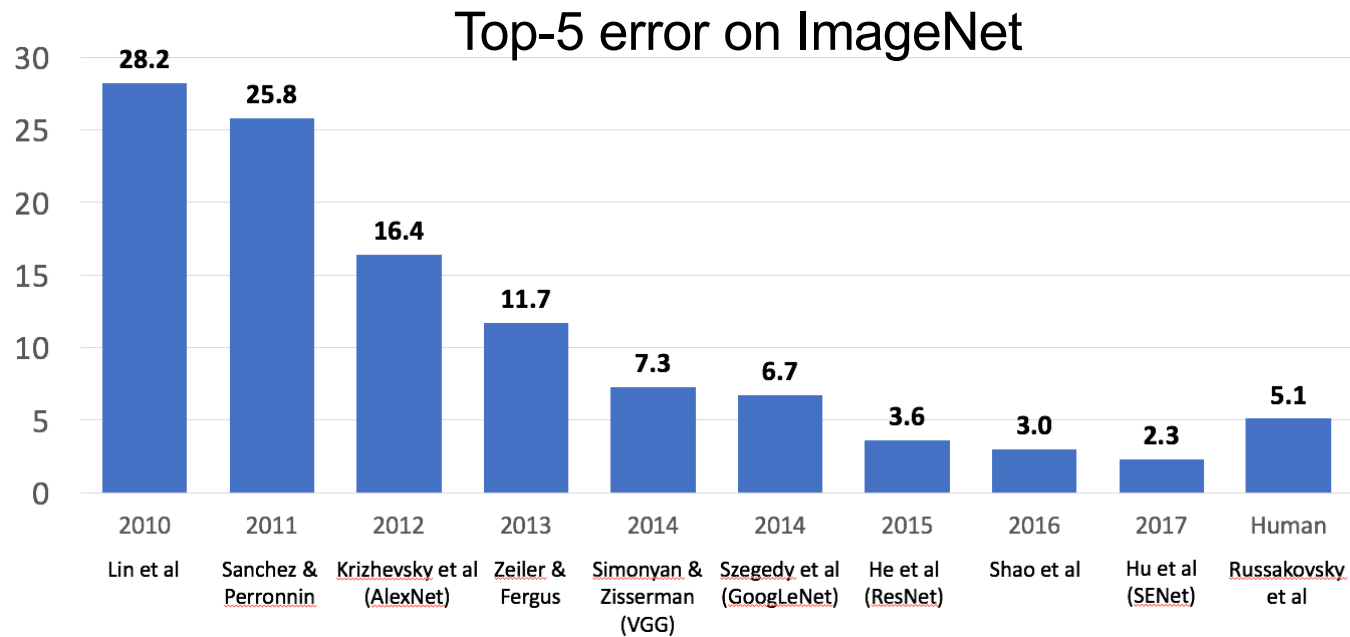
- Intellectually interesting
 - We are too good at it to appreciate how difficult it really is.
 - Going from 2D to 3D is an under-determined inverse problem.
- Psychology
 - After all, about 50% of cerebral cortex is for vision.
 - Vision is (to a large extent) how we experience the world.
- Engineering
 - It is fun to build "intelligent" machines.
 - Computer vision opens up for multi-disciplinary work.
 - We have images everywhere and need ways to organize them.

The growth of computer vision



Number of submitted (blue) and accepted (green) papers at CVPR

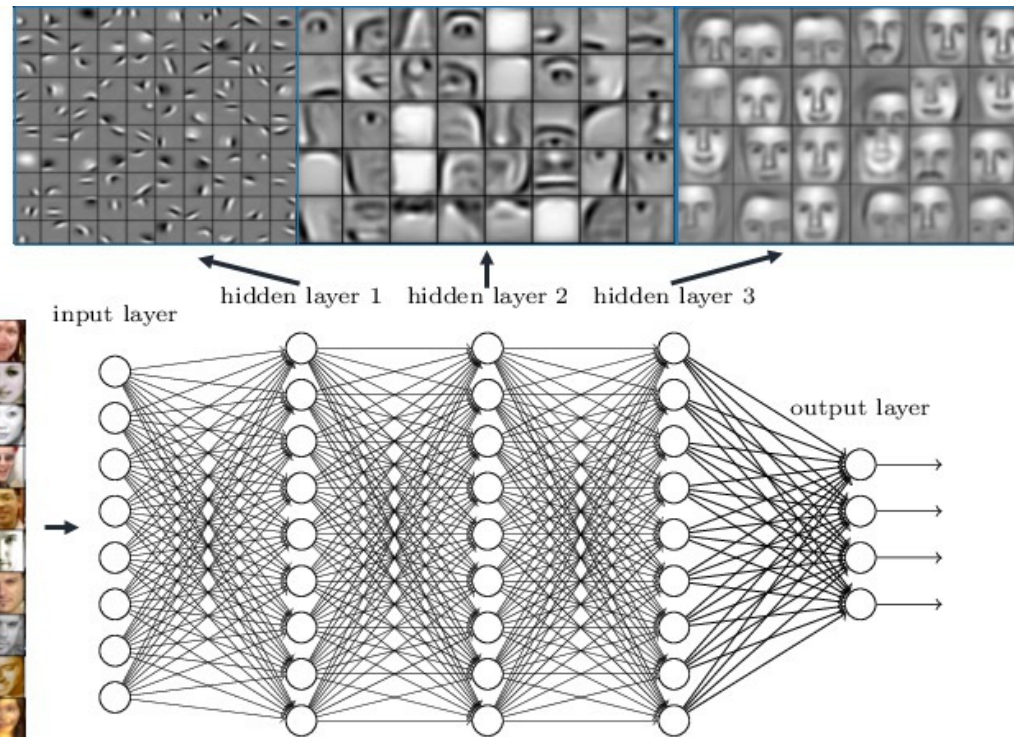
Deep networks for image classification



- Neural networks were long forgotten in computer vision.
- But... deep neural networks have become state-of-the-art.
- The best networks today have an error rate of about 1%.

Fully Connected Neural Network (FCN)

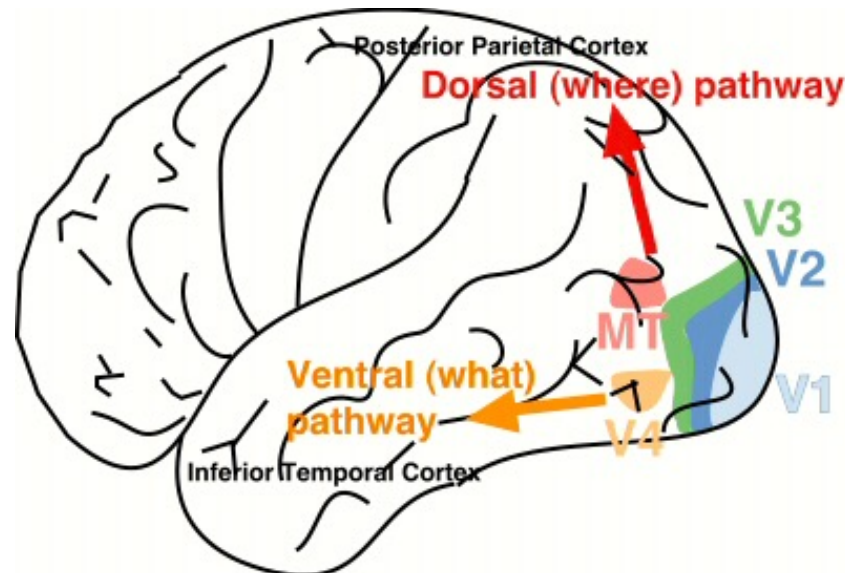
Deep neural networks learn hierarchical feature representations



- Neural networks typically consists on layers of neurons.
- Possibly images on inputs, some hypotheses on outputs.

What about deep learning?

Visual cortex with *what* and *where* pathways

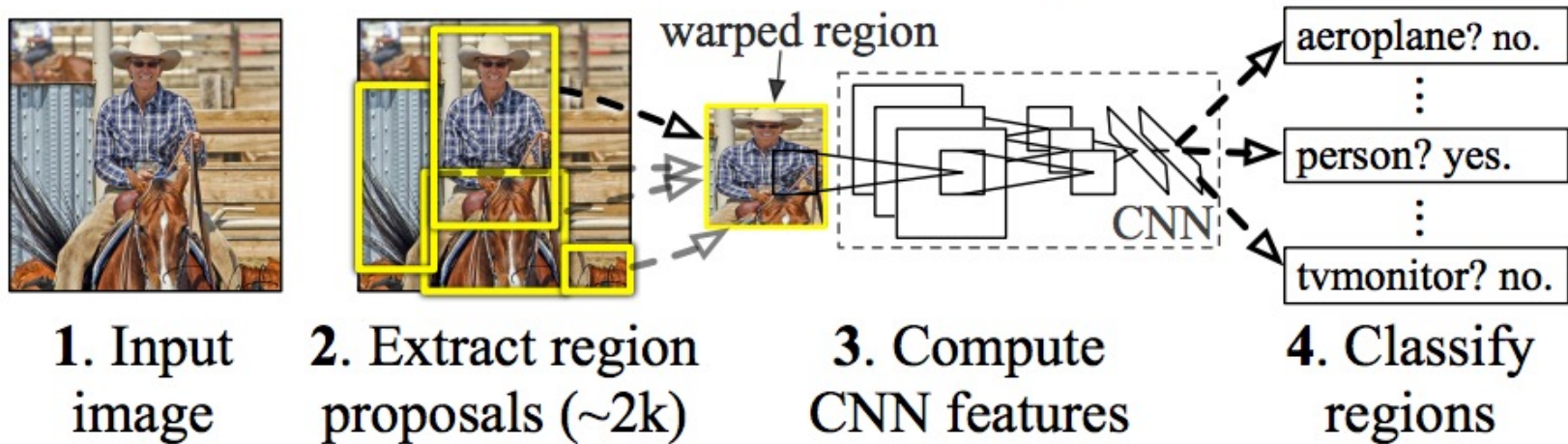


Deep learning can

- answer *what*-questions – but *where*-questions are also important.
- benefit from lots of data – but what if you don't have much data?

Object detection with CNN

R-CNN: *Regions with CNN features*



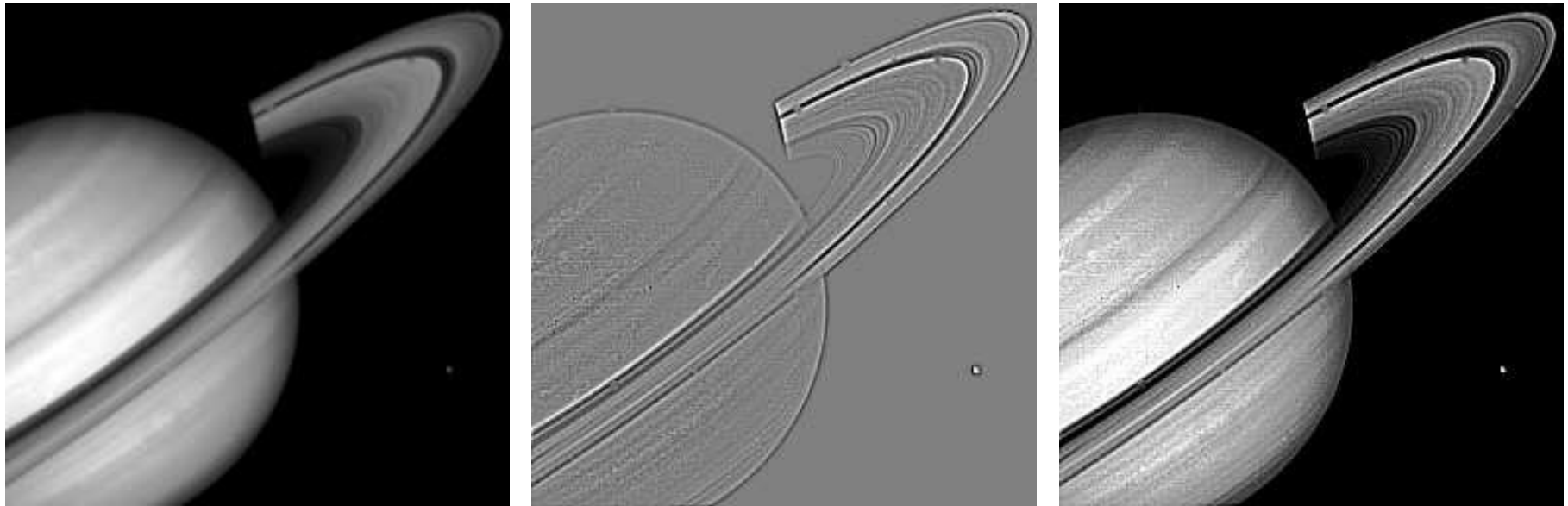
A *where*-problem essentially solved by through many *what*-questions.

Image processing $\Leftrightarrow$ Signal processing



- The image is enhanced for easier interpretation.
- Different levels of processing (often used as pre-processing).

Example: Laplacian filter for sharpening



Original image (left), application of Laplacian operator (middle), and subtraction of the Laplacian from the original image (right).

Example: Anisotropic smoothing



- Blur an image to remove noise, but control in which direction to blur.
- Note: the image is smoother, but individual hairs are not blurred out.



Image analysis



Processing



Compact
description

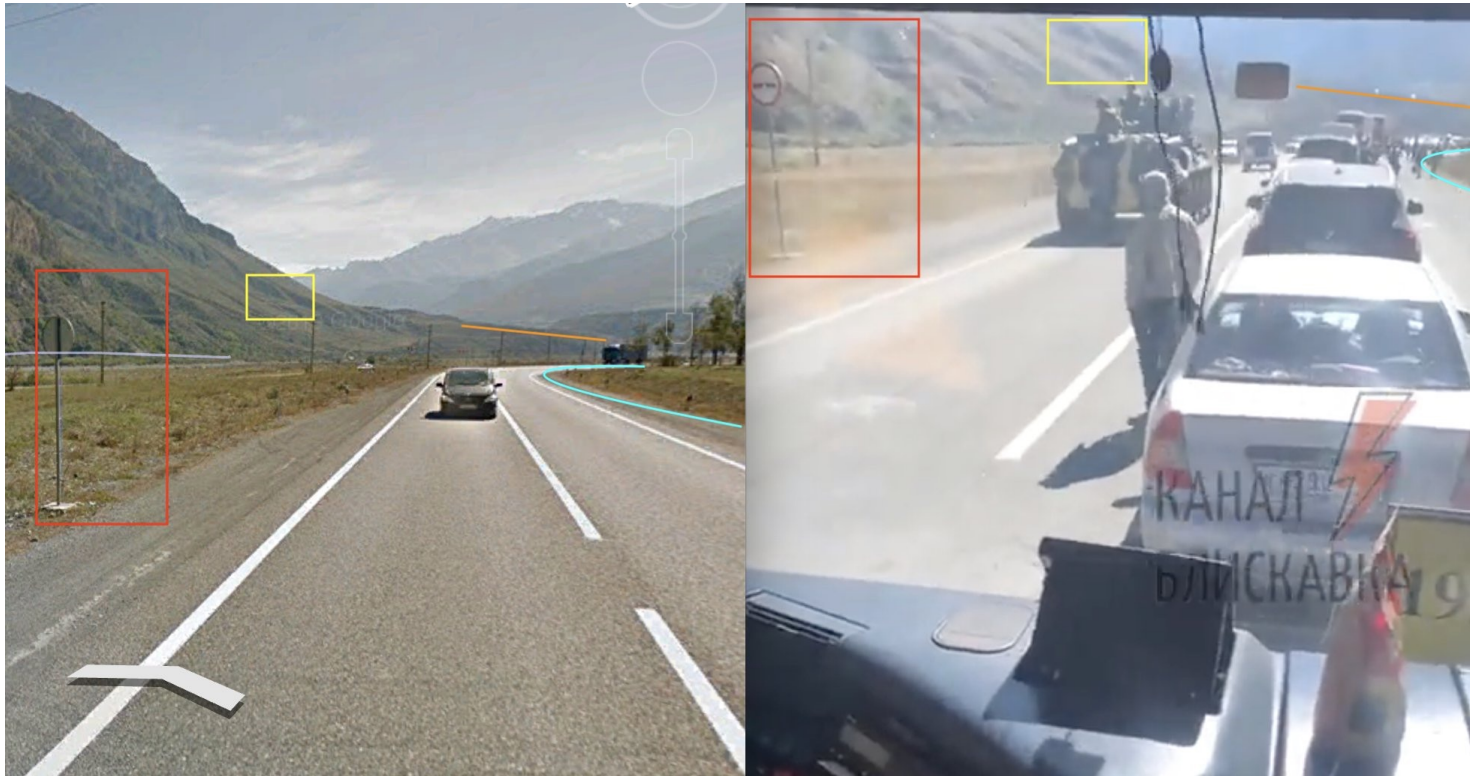
- Purpose: Generate a useful description of the image
- Feature detection and matching
- Object and image segmentation
- Recognition and classification

Feature extraction and matching



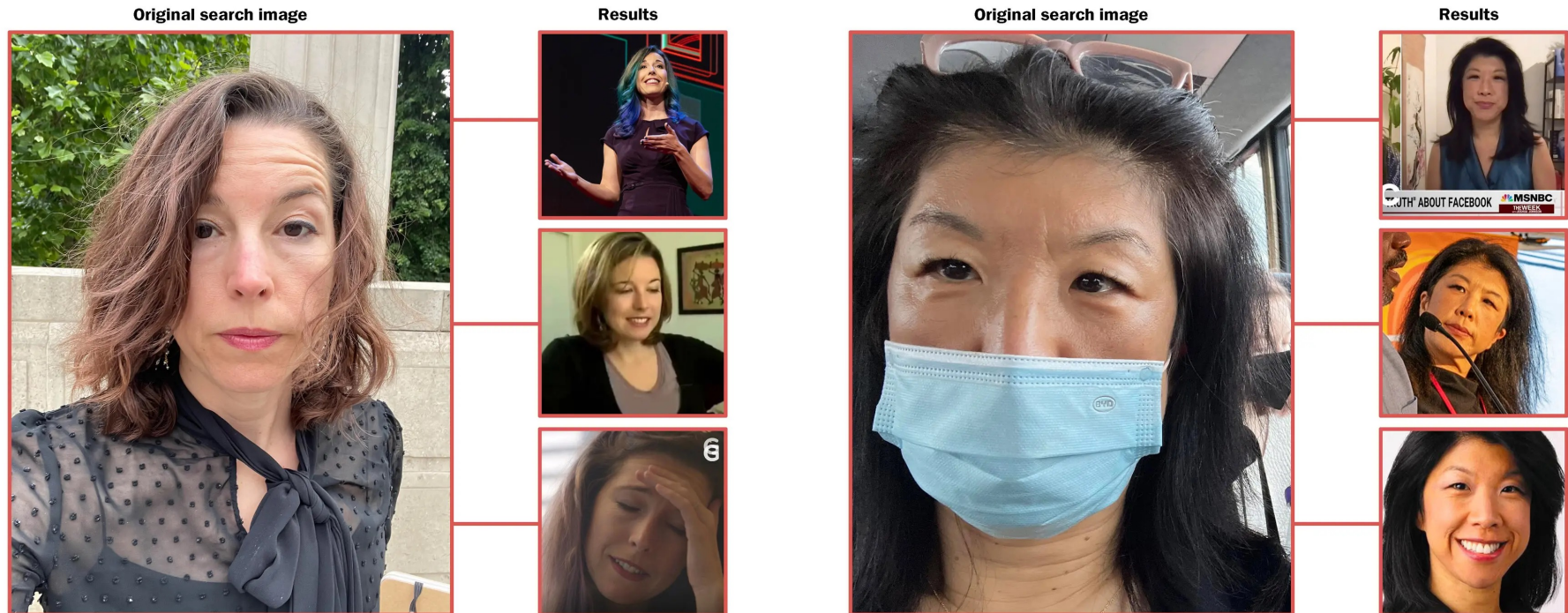
- Features (corners, blobs, lines, etc) can be extracted from images and then matched between images.

Example: Geolocalisation



Confirmation of Russian troops moving towards the border of Georgia. September 26, 2022.

Example: Reverse face recognition – scary?



Given a single image, images of the same person found on Internet using PimEyes (New Your Times)

Computer vision



- Purpose: Achieve an understanding of the world, possibly under active control of the image acquisition process.
- Examples: object tracking, activity recognition
- Often people say computer vision, instead of image analysis.

Example: Object segmentation using GrabCut



- Left: A couple of strokes applied to create colour models of background and foreground.
- Right: Afterwards background can be changed to something else.

Example: Semantic segmentation



If you know what a scene usually consist of, you can exploit that for image segmentation.

Example: Google self-driving cars



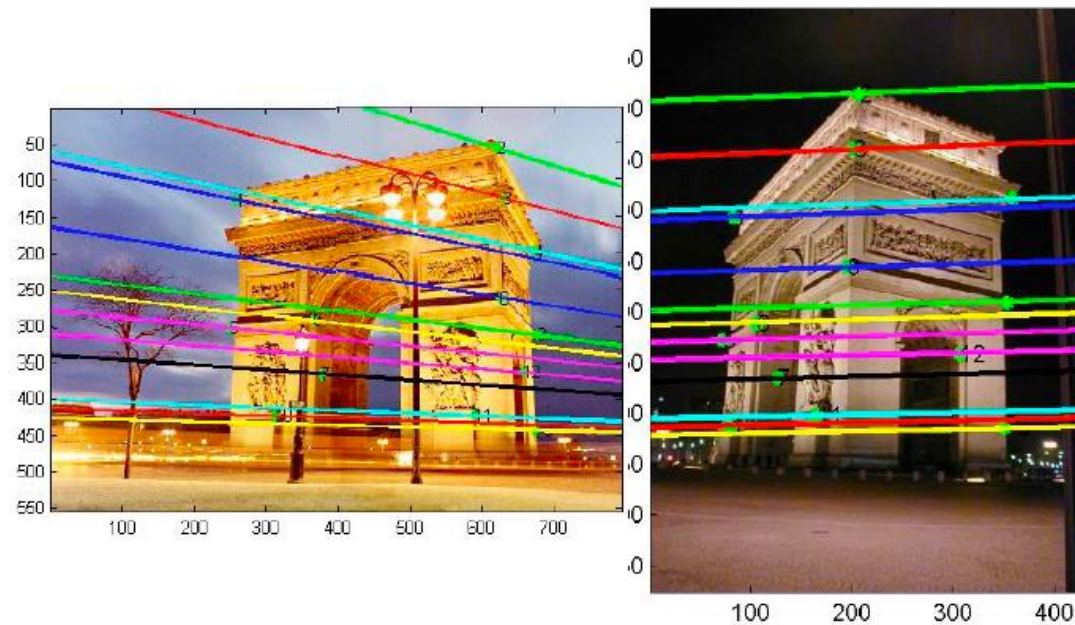
This is what a Google self-driving car is seeing. Hopefully, it has been improved since then.

Example: Active contours



If you segment a particular object, just track the contours around the object.

Epipolar geometry for stereo matching

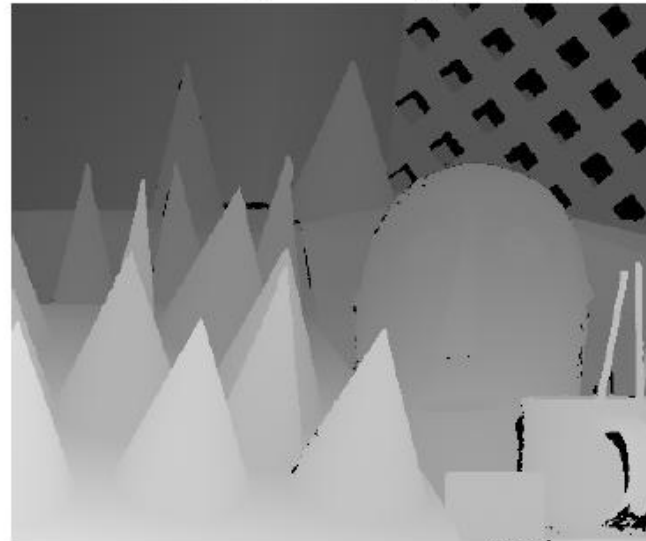


- Matches between two images can be found along lines determined by the relative positions between cameras, or vice versa,
- If you have a number of matches between images, you can determine the relative positions between cameras.

Stereo matching



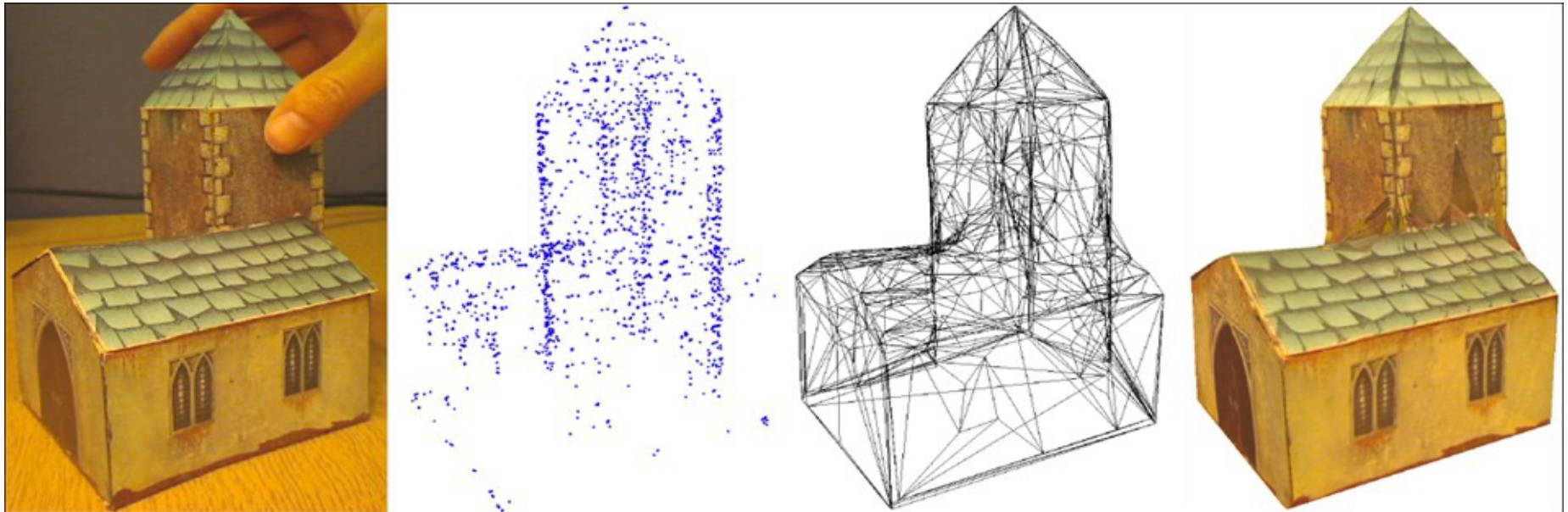
Disparity values (0-64)



Note how disparity is larger (brighter) for closer surfaces.

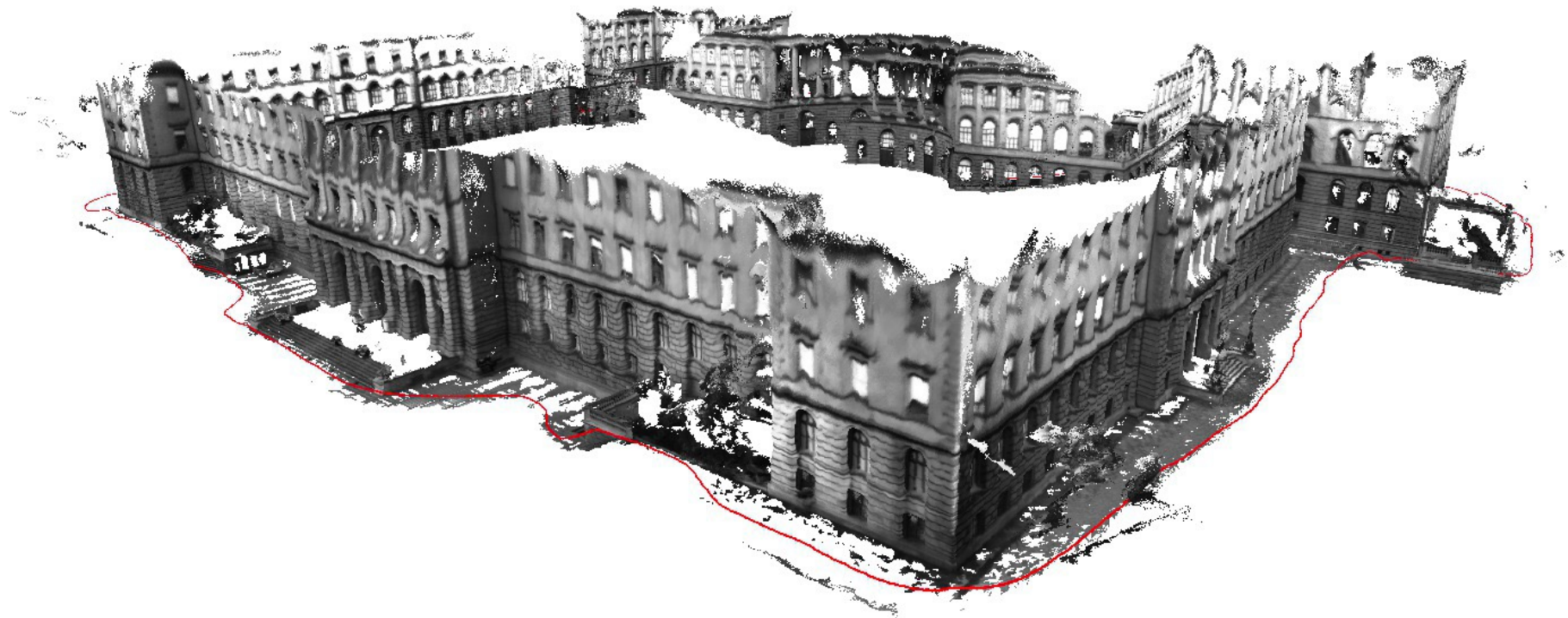
- Match left and right images to create a depth image. The problem is how to benchmark your method, since then you need some ground truth data.

Multi-view 3D object reconstruction



Find image features in multiple views, determine their positions in the 3D world and create surfaces.

Multi-view 3D object reconstruction



Such models can be very large and used to control a robot.



NeRF: Neural Radiance Fields (UC Berkeley)

Synthetic Scenes

Model (colour, density) as a function of (position, direction) with a neural network.

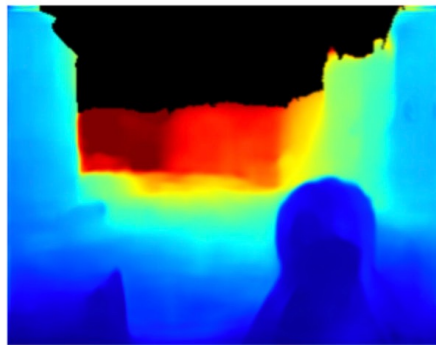


NeRF: Neural Radiance Fields

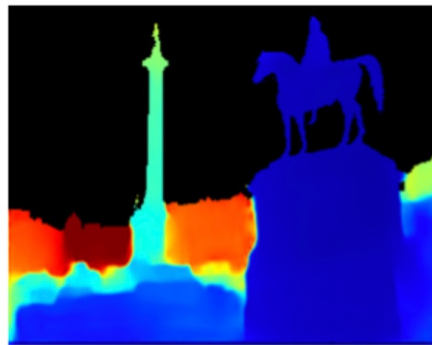


From a couple of mobile phone images matched with `colmap` and 4 hours of training.

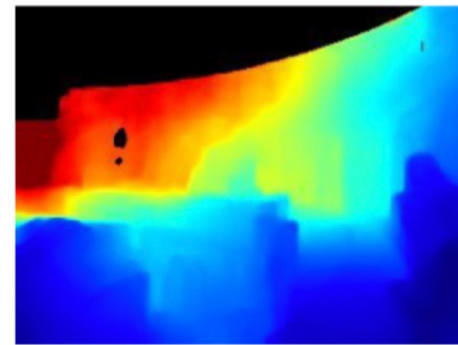
Single-view depth using deep learning



Grand Canal, Venice



Trafalgar Square, London



Colosseum, Rome

With deep learning one can predict depths from a single view, but results are not perfect.

Motion analysis



Measure how pixels move over time. Similar to stereo, but more complicated.

Example: Multi-person tracking (CMU)



If you know you are tracking human bodies, the known shape of the body can be exploited.



Thank you!









